SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I(Devadharshini E / 192524483) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Devadharshini E

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Selected Peer-Reviewed Research Article

Title	Logistic regression was as good as machine learning for predicting major chronic diseases
Authors	Simon Nusinovici, Yih Chung Tham, Marco Yu Chak Yan, Daniel Shu Wei Ting, Jialiang Li, Charumathi Sabanayagam, Tien Yin Wong, and Ching-Yu Cheng
Journal	Journal of Clinical Epidemiology
Volume / Pages	Volume 122, June 2020, pp. 56–69
DOI	10.1016/j.jclinepi.2020.03.002
Study Type	Comparative prospective cohort study

- Source note: The article compares standard logistic regression with five machine-learning models (single-hidden-layer neural network, support vector machine, random forest, gradient boosting machine and k-nearest neighbor) for predicting cardiovascular disease, chronic kidney disease, diabetes and hypertension using simple clinical predictors in a population-based cohort.

Task 1: Article Summary

(a) Full citation and significance

- Nusinovici, S., Tham, Y. C., Chak Yan, M. Y., Ting, D. S. W., Li, J., Sabanayagam, C., Wong, T. Y., & Cheng, C. Y. (2020). Logistic regression was as good as machine learning for predicting major chronic diseases. Journal of Clinical Epidemiology, 122, 56–69. https://doi.org/10.1016/j.jclinepi.2020.03.002.
- This article is highly relevant to the Artificial Intelligence course because it directly examines whether more complex machine-learning algorithms actually provide a meaningful predictive advantage over a classical statistical-learning method. The study is especially useful for understanding model selection, evaluation, interpretability and the trade-off between complexity and predictive performance.

(c) Summary of the article

- The study investigates whether machine-learning methods outperform conventional logistic regression when predicting major chronic diseases from relatively simple clinical variables. The authors used data from 6,762 Asian adults in a population-based prospective cohort. Five machine-learning approaches were compared with standard logistic regression: a single-hidden-layer neural network, support vector machine, random forest, gradient boosting machine and k-nearest neighbor.

- The outcomes studied were cardiovascular disease (CVD), chronic kidney disease (CKD), diabetes mellitus (DM) and hypertension (HTN). At approximately six years, the observed incidences were 4.0% for CVD, 7.0% for CKD, 9.2% for diabetes and 34.6% for hypertension. Logistic regression produced the highest AUC for CKD (0.905) and diabetes (0.768). For CVD, the best model was the neural network with AUC 0.753, while for hypertension the support vector machine had the highest AUC at 0.780. However, the differences from logistic regression were small and statistically non-significant.
- The central conclusion is that logistic regression can perform as well as more complex machine-learning methods when the dataset has moderate size, a limited number of events and a relatively small set of simple clinical predictors. Therefore, greater algorithmic complexity should not automatically be assumed to produce better clinical prediction.

Research Gap Identified

- The study focuses mainly on static risk prediction. It does not address the problem of sequential treatment decisions in which a patient's condition changes over time and an intelligent agent must select actions based on previous actions and observed outcomes. It also relies on a relatively limited set of clinical predictors. The research can therefore be extended toward richer longitudinal electronic-health-record data, external validation across healthcare settings, calibration and fairness analysis, and reinforcement-learning methods for sequential decision support.
- Proposed gap-filling direction: combine interpretable statistical risk prediction with

a carefully constrained reinforcement-learning layer that operates only on validated longitudinal data and recommends candidate actions for clinician review rather than making autonomous treatment decisions.

Task 2: Methodology Analysis

(a) Machine-learning techniques used in the article

Technique	Category	Role
Logistic Regression	Statistical learning / binary risk prediction	Provides a probability of an outcome and is comparatively interpretable.
Single-hidden-layer Neural Network	Supervised machine learning	Learns nonlinear relationships using a hidden layer.
Support Vector Machine	Supervised machine learning	Finds a separating boundary with maximum-margin principles.
Random Forest	Ensemble decision-tree learning	Combines multiple trees to reduce variance and improve robustness.
Gradient Boosting Machine	Ensemble / boosting	Builds models sequentially so later learners focus on previous errors.
K-Nearest Neighbor	Instance-based supervised learning	Predicts using outcomes of nearby observations in feature space.

(c) Mapping to CO4 topics

CO4 Topic	Connection to Article
Inductive Learning	The models learn general predictive patterns from labelled cohort data and apply them to unseen observations.
Decision Trees	Random Forest is directly based on decision trees; individual trees recursively split data

	using predictor values.
Statistical Learning	Logistic regression is the central classical statistical model used as the benchmark.
Reinforcement Learning	Not a primary method in the article. It is a suitable extension because the paper studies prediction, while RL addresses sequential decisions.

(c) Strengths and limitations of the methodology

- Strengths:
- Uses a prospective population-based cohort rather than a purely synthetic dataset.
- Compares several machine-learning approaches against a conventional statistical baseline.
- Evaluates multiple clinically important chronic diseases rather than one outcome.
- Uses AUC as a discrimination measure, allowing comparison across models.
- The comparison supports a practical lesson: model complexity should be justified by measurable improvement.
- Limitations:
- The predictors are relatively simple, so the findings may not generalize to high-dimensional EHR, imaging or multimodal data.
- The study is primarily concerned with prediction rather than sequential treatment optimization.
- Performance in one population may not transfer directly to other healthcare systems or demographic groups.
- Low-incidence outcomes can make model training and comparison difficult because the number of positive events is limited.
- Clinical deployment would require strong external validation, calibration assessment and prospective evaluation.

Task 3: Results & Critical Evaluation

(a) Key reported metrics

Outcome	**Incidence at ~6 years**	**Best reported model**	**AUC**	**Interpretation**
CVD	4.0%	Neural Network	0.753	Best model only slightly ahead of logistic regression.
CKD	7.0%	Logistic Regression	0.905	Logistic regression achieved the highest AUC.
Diabetes	9.2%	Logistic Regression	0.768	Logistic regression achieved the highest AUC.
Hypertension	34.6%	Support Vector Machine	0.780	Difference from logistic regression was small.

- The article's results support the authors' conclusion that logistic regression was not meaningfully outperformed by the tested machine-learning models for the studied setting. In particular, logistic regression had the highest AUC for CKD and diabetes, while the differences for CVD and hypertension were small and statistically non-significant.

(b) Critical evaluation of whether the metrics are realistic

- The reported AUC values are realistic for clinical risk prediction because AUC measures discrimination rather than guaranteeing perfect individual diagnosis. The values also differ by disease, which is expected because different diseases have different event rates and predictor strength. However, AUC alone is insufficient for clinical deployment. A model can have a reasonable AUC while being poorly calibrated or producing unacceptable false-negative rates at the chosen decision threshold.

- For a stronger evaluation, the study or a follow-up implementation should report sensitivity, specificity, precision, F1-score, calibration plots, Brier score, decision-curve analysis and confidence intervals, together with external validation on an independent population.

(c) Potential biases and limitations

- Selection bias: the cohort population may differ from patients in other regions, hospitals or age groups.
- Spectrum bias: predictors and disease prevalence may not represent all levels of disease severity.
- Class imbalance: low-incidence diseases provide fewer positive events, making recall and precision more difficult to estimate robustly.
- Measurement bias: clinical predictors can contain errors, missing values or differences in measurement procedures.
- Dataset shift: healthcare practice, population characteristics and diagnostic criteria may change over time.
- Fairness risk: a model that performs well on the overall cohort may perform differently across demographic subgroups.
- External-validity limitation: strong internal performance does not automatically mean safe deployment in another institution.

(d) Real-world applicability

- The findings are highly relevant to clinical decision-support systems because they show that a simpler model can be competitive when the number of predictors is limited. Logistic regression is attractive where clinicians need probability estimates, transparent coefficients and lower computational complexity. More complex machine-learning models become more attractive when the data contain nonlinear interactions, high-dimensional EHR variables, images, signals or longitudinal information.

- Scaling to industry requires reliable data pipelines, secure storage, interoperability with electronic health records, monitoring for data drift, model recalibration, privacy protection, audit logs and clinical governance. The model should support—not replace—clinical judgment.

Task 4: Reflection & Learning Outcome

(a) Three key insights gained

- **Model complexity is not the same as model quality.** A sophisticated algorithm is useful only when it provides meaningful improvement over a simpler baseline.
- **Evaluation must match the clinical objective.** AUC is useful, but clinical decisions also require sensitivity, specificity, calibration, subgroup analysis and threshold-based assessment.
- **Explainability and feasibility matter.** A transparent model such as logistic regression may be preferable when its predictive performance is comparable to complex alternatives.

(b) Proposed extension: Reinforcement Learning for Sequential Clinical Decision Support

- A feasible extension is to build a longitudinal clinical decision-support framework in which a patient's condition is represented as a state and the system learns which candidate intervention is associated with better long-term outcomes. The reinforcement-learning agent would not directly prescribe treatment. Instead, it would generate a ranked recommendation for clinician review.

MDP Component	Proposed Design
State	Patient risk score, recent laboratory values, vital

	signs, previous outcomes and treatment history.
Actions	Candidate clinical-management actions such as monitoring, lifestyle intervention or clinician-reviewed treatment options.
Reward	Validated clinical outcome improvement, with strong penalties for adverse events and unsafe recommendations.
Transition	Change in the patient's health state between successive clinical visits or time windows.
Policy	A mapping from patient state to the action with the best expected long-term outcome.

- Feasibility: longitudinal EHR datasets can provide repeated observations, actions and outcomes, making them potentially suitable for offline RL research. Expected impact: the extension could move from static risk prediction toward personalized sequential decision support.
- However, offline RL is challenging because historical clinical actions are confounded by clinician judgment, rare actions may have insufficient examples, and unsafe policies cannot be tested directly on patients. Therefore, simulation, conservative/off-policy evaluation and strict clinical oversight are essential.

Proposed Improved Architecture

- Patient/EHR Data
- Data Cleaning + Feature Engineering
- Baseline Logistic Regression / Decision Tree
- Risk Prediction + Calibration
- Longitudinal State Representation
- Offline Reinforcement Learning
- Safety Constraints + Bias Checks

- Clinician Review
- Outcome Monitoring + Model Recalibration

Overall Conclusion

- The reviewed article demonstrates that logistic regression can match or exceed several machine-learning methods for chronic-disease risk prediction when the data are moderate in size, events are relatively limited and predictors are simple. Its strongest lesson is that an AI system should be selected on evidence rather than algorithmic complexity.
- For future work, the static prediction framework can be extended with longitudinal data and carefully constrained reinforcement learning to study sequential clinical decision support. Such an extension must emphasize external validation, fairness, calibration, privacy, explainability and human clinical oversight.

References

- Nusinovici, S., Tham, Y. C., Chak Yan, M. Y., Ting, D. S. W., Li, J., Sabanayagam, C., Wong, T. Y., & Cheng, C. Y. (2020). Logistic regression was as good as machine learning for predicting major chronic diseases. Journal of Clinical Epidemiology, 122, 56–69. https://doi.org/10.1016/j.jclinepi.2020.03.002.
- Nusinovici, S. et al. (2020). PubMed record for the selected article, PMID 32169597.
- Christodoulou, E. et al. (2019). A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models. Journal of Clinical Epidemiology, 110, 12–22.